

EEG-Informed Adaptive User Experience Questionnaire Generation with Multi-Agent Large Language Models



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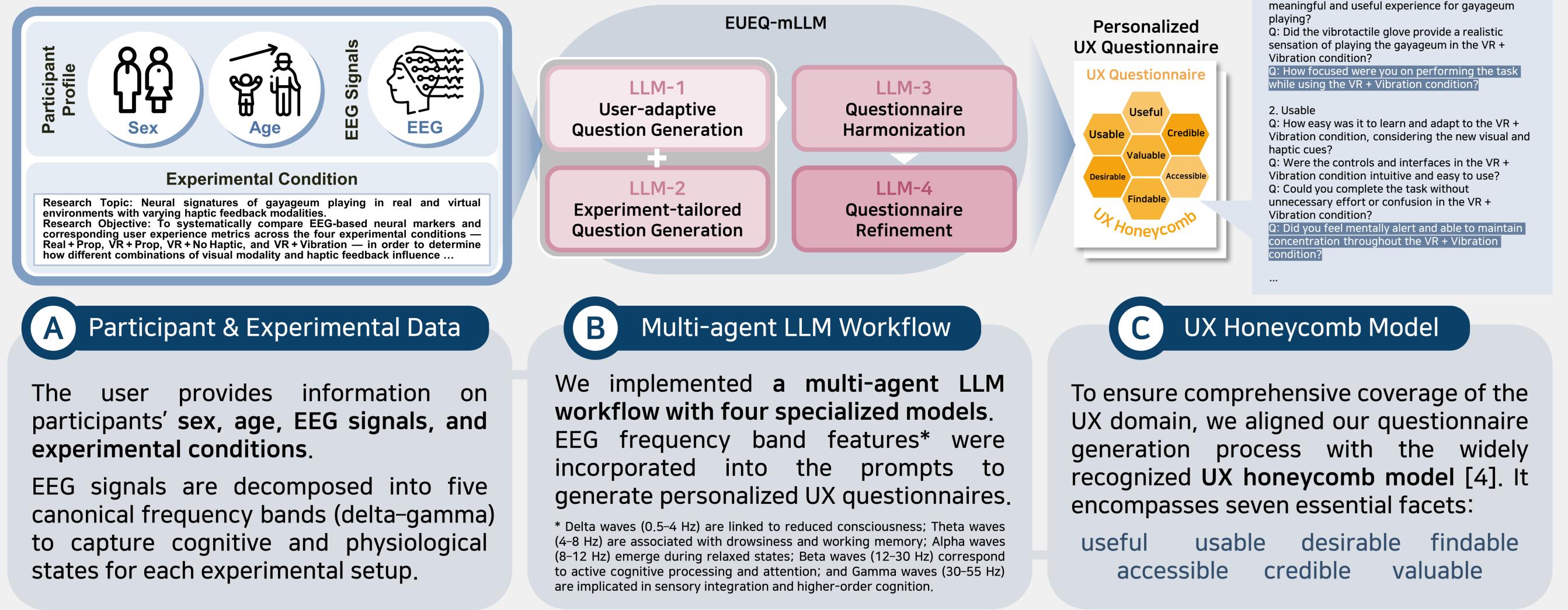
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Introduction

- Immersive technologies such as virtual reality and haptic systems create complex multisensory experiences that challenge traditional user experience (UX) evaluation methods [1, 2].
- Conventional static questionnaires fail to capture individual variability and the dynamic nature of user responses, and interpreting electroencephalography (EEG) signals for UX evaluation remains challenging to use in practice [3].
- This study proposes **EUEQ-mLLM**, a novel framework for EEG-informed adaptive User Experience Questionnaire generation using multi-agent Large Language Models (LLMs).
- EUEQ-mLLM** employs multiple specialized LLMs that collaboratively generate UX questionnaires informed by EEG frequency-band characteristics and experimental contexts, enabling personalized and physiologically grounded assessment of user experience.

Methods



Experiments

- EEG signals were recorded from participants performing a musical task under four visuo-haptic conditions, using a 21-channel headset and standard preprocessing to extract frequency band features for analysis.
- To establish a benchmark for evaluating the performance of the **EUEQ-mLLM**, we implemented a single LLM-based approach, denoted as **EUEQ-sLLM**. This baseline model consolidated the functionalities of the four LLMs in the agentic workflow into a single LLM instance.
- To generate personalized UX questionnaires, we used Meta's open-source LLM, *Meta-Llama-3.1-8B-Instruct* [5].
- We adapted the G-Eval framework to evaluate the quality of the generated questionnaires. Four primary dimensions were established: **consistency**, **personalization**, **fluency**, and **relevance** [6].
- We utilized OpenAI's *GPT-5* as the underlying LLM for G-Eval with parameters set as *temperature* = 1, *top_p* = 1, and reasoning effort specified as "medium."

Results

- G-Eval scores were calculated for each questionnaire, with all items rated on a 5-point scale.
- To facilitate comparison, scores were normalized to a 100-point scale.

Criteria	EUEQ-mLLM	EUEQ-sLLM	EUEQ- mLLM vs. EUEQ-sLLM	
			t	P
Consistency	4.13	1.43	10.47	< 0.001
Fluency	4.79	4.90	-1.30	0.200
Personalization	3.52	2.54	5.51	< 0.001
Relevance	3.76	2.17	18.82	< 0.001
Score	80.96	55.71	-	-

The EUEQ-mLLM consistently outperformed the EUEQ-sLLM

EUEQ-mLLM achieved a score of 80.96, representing a 45.32% improvement over EUEQ-sLLM, which has a score of 55.71.

These results demonstrate that a multi-LLM approach is effective for generating high-quality, user-adaptive UX questionnaires.

Conclusions

- This study introduced **EUEQ-mLLM**, a novel framework designed to generate adaptive UX questionnaires by integrating EEG signals with multi-agent LLMs.
- The proposed methodology offers a promising approach to enhancing UX evaluation by providing real-time insights into user behavior and preferences.

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